

Problem Statement

Given a customer's profile, banking relationship, transactional behavior, digital engagement, complaint history, and marketing response data, predict whether they will churn (close their account or become inactive) in the upcoming period.

Dataset

Two files are provided: ChurnZero_Dataset_v1.csv (8,101 rows, target included) and ChurnZero_Test_v1.csv (2,026 rows, no target). 97 features across 8 categories: customer profile, relationship & tenure, account & transaction behavior, product holding, credit card & loan behavior, digital banking engagement, service & complaints, and marketing & retention.

Category	Example Features
Customer Profile (12)	age, gender, marital_status, education_level, occupation, annual_income, city_tier,
Relationship & Tenure (10)	tenure_months, onboarding_channel, number_of_products, customer_lifetime_value,
Account & Transaction Behaviour (15)	avg_monthly_balance, balance_decline_percentage, monthly_transaction_count,
Product Holding (10)	savings_account_flag, credit_card_flag, home_loan_flag, fixed_deposit_flag, investment_product_flag,
Credit Card & Loan Behaviour (10)	credit_card_limit, credit_utilization_ratio, late_credit_card_payment_count, loan_default_risk_score
Digital Banking Engagement (10)	mobile_app_login_count, digital_transaction_ratio, last_login_days, email_open_rate
Service & Complaint Behaviour (10)	total_complaints, complaint_resolution_time, satisfaction_score, nps_score, escalation_count, ...
Marketing & Retention (10)	campaign_received_count, retention_offer_accepted, competitor_bank_offer_awareness

Deliverables

Submit one zip file named ChurnZero_<TeamName> containing exactly three files:

File	What it must contain
ChurnZero_<TeamName>_Predictions.csv	The CSV file will contain churn_prediction (0/1) and churn_probability (float, 0-1) columns. The file will have exactly 2,026 rows with matching customer_id values and no nulls in the new columns.
ChurnZero_<TeamName>_Presentation.pptx / pdf	Slide deck for a non-technical exec audience. Cover EDA, feature engineering, model choice, results (PR-AUC, F1, confusion matrix, business cost), top churn drivers, and recommendations. Max 15 slides including first and last.
ChurnZero_<TeamName>_Code	Reproducible Python/R notebook. Run end-to-end. Accepted: .ipynb, .py, .Rmd, .R.

Evaluation Metrics

Component	Weight	Notes
Model Performance	40%	Primary metric: PR-AUC on held-out test labels. Secondary: F1 on positive class. Business-cost framing: FN cost ₹40,000, FP cost ₹500.
Presentation Quality	25%	Clarity for an exec audience, depth of insight, business framing.
Methodology & Rigour	20%	Feature engineering decisions and ensure no data leakage.
Code Quality	15%	Reproducibility, readability, organization, and comments.